SmartManageAI

Complete Platform Feature & Integration Guide

Enterprise-grade, white-label SaaS platform for the construction industry — from bid to closeout.

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Platform Overview

SmartManageAI is an enterprise-grade, white-label SaaS platform purpose-built for construction companies of every size and trade. It centralizes the entire construction project lifecycle into one intelligent system — from winning bids to final closeout — while providing AI-driven insights, real-time collaboration, and trade-specific customization for over 230 construction trades across 17 categories. The platform is multi-tenant by design, meaning each organization gets a fully isolated, brandable environment with its own users, data, and configuration.

1. AI-Powered Intelligence Suite

SmartManageAI embeds artificial intelligence into every layer of the platform, powered by OpenAI's GPT-4o and GPT-4o Vision models. These are not add-on gimmicks — they are deeply integrated tools that learn from your data and produce actionable results.

AI Project Manager (AI PM)

The AI Project Manager acts as a virtual superintendent, continuously analyzing every active project to surface risks, generate tasks, and recommend actions before problems escalate.

- Automatically scans project data including budgets, labor hours, material consumption, schedule progress, and weather forecasts.
- Generates prioritized task lists with urgency ratings (Critical, High, Medium, Low).
- Produces risk alerts when budget overruns, schedule delays, or safety concerns are detected.
- Runs on-demand or on a scheduled cadence, delivering insights directly to the dashboard.
- Maintains a full history of AI-generated analyses for comparison and accountability.

AI CFO & Financial Analyst

A dedicated financial intelligence engine that gives owners and executives instant visibility into the financial health of every project and the business overall.

- Calculates project profitability using real-time data from labor, materials, subcontracts, and change orders.
- Identifies cost variances and flags projects where spending is trending above estimates.
- Provides cash flow forecasting based on Schedule of Values billing, accounts receivable, and payable commitments.
- Generates executive-level financial summaries with plain-language explanations of complex financial data.

AI Auto-Takeoff (Computer Vision)

One of the platform's most powerful features, the AI Auto-Takeoff uses computer vision to analyze uploaded construction blueprints and automatically extract material quantities.

- Accepts PDF plans and high-resolution images as input.
- Uses GPT-4o Vision to identify and measure linear distances, areas, and item counts directly from the drawings.
- Results are displayed on the interactive canvas for review, adjustment, and approval.
- Dramatically reduces the time spent on manual quantity takeoffs from hours to minutes.

AI Scheduling Assistant

Takes the guesswork out of project scheduling by generating optimized timelines based on your actual project data.

- Analyzes scope of work items, historical production rates, and crew availability to propose schedules.
- Factors in weather forecasts and identifies potential weather-related delays.
- Highlights critical path tasks and scheduling conflicts.
- Produces interactive Gantt charts that can be manually adjusted after AI generation.

AI Bid & Contract Analysis

Helps estimators and project managers quickly understand the requirements and risks in complex bid packages and contracts.

%ï Extracts key information from bid documents including dates, scope requirements, and special conditions.
%ï Analyzes contracts for liability clauses, payment terms, insurance requirements, and risk flags.
%ï Highlights unusual or potentially problematic contract language.
%ï Summarizes multi-page documents into actionable bullet points.

AI Email Assistant & Classifier

Turns your inbox into a managed workflow by automatically understanding and categorizing incoming communications.

%ï Classifies emails into categories: RFI, Change Order Request, Billing Question, Delay Notice, Technical Question, and more.
%ï Extracts actionable items and can automatically create tasks from email content.
%ï Generates draft replies with selectable tone (Professional, Casual, Firm).
%ï Provides daily and weekly email digest summaries.

AI Change Order Generator

Streamlines the change order process by automatically drafting professional change order documents.

%ï Input a description of the scope change, and the AI generates a complete change order document with justification, impact assessment, and cost implications.
%ï Pulls relevant project data to support the change order narrative.
%ï Produces documents ready for review and client presentation.

AI Jobsite Camera Analysis

Brings computer vision to the field by analyzing jobsite photographs for progress, quality, and safety observations.

%ï Upload site photos and receive AI-generated observations about construction progress, material staging, and work quality.
%ï Identifies potential safety concerns visible in site imagery.
%ï Helps document daily progress with AI-written summaries tied to photographic evidence.

AI Website Wizard

A guided AI assistant that helps construction companies build professional, SEO-optimized websites (see Website Builder section for full details).

AI Scope Prediction

Predicts the scope of work items a project will need based on trade type, project characteristics, and historical data from similar projects, accelerating the estimating process.

AI Vendor Pricing Extraction

Automatically extracts pricing data from vendor PDF catalogs, Excel spreadsheets, and price lists, populating your pricebook without manual data entry.

AI Onboarding & Trade Discovery

A wizard that researches specific construction trades to pre-configure the platform with industry-standard scope items, materials, formulas, and calculators, so new users are productive from day one.

AI Governance Dashboard

Maintains transparency and control over AI usage across the organization.

- Tracks AI token usage, costs, and response quality.
- Provides a "Human-in-the-Loop" verification process for all AI-generated content.
- Logs all AI interactions for compliance and audit purposes.
- Allows administrators to configure AI behavior and access permissions.

2. Preconstruction & Bidding

Bid Management

A complete bid tracking system that manages the entire lifecycle from invitation to award.

- %ï Track all active, pending, and awarded bids in a centralized dashboard.
- %ï Capture bid details including project name, client information, due dates, and estimated values.
- %ï Monitor bid status progression: Draft, Submitted, Under Review, Won, Lost.
- %ï Attach bid documents, plans, and specifications directly to bid records.
- %ï Calculate win rates and pipeline values for business forecasting.

AI Bid Wizard

A guided, AI-powered workflow for processing new bid packages.

- %ï Upload bid documents and the AI extracts project information, scope requirements, deadlines, and special conditions.
- %ï Automatically creates a structured bid record with all relevant details populated.
- %ï Suggests scope items and quantities based on the bid package analysis.
- %ï Reduces bid processing time from hours to minutes for complex packages.

Bid Network (GC/Subcontractor Collaboration)

A built-in network for General Contractors and Subcontractors to collaborate on bids.

- %ï GCs can send structured bid invitations to subcontractors for specific scopes of work.
- %ï Subcontractors receive, review, and submit structured bid responses.
- %ï Supports clarification questions and answers between parties.
- %ï Automatic email notifications for new invitations, responses, and clarifications.
- %ï Track invitation status: Sent, Viewed, Responded, Accepted, Declined.

Bid Inbox

A centralized inbox for managing incoming bid invitations and opportunities.

- %ï View all incoming bid requests from general contractors.
- %ï Quick-respond to opportunities with structured pricing and scope confirmations.
- %ï Filter and sort by due date, trade, and value.

Bid Document Distribution

Manage the distribution of bid documents to the right team members and subcontractors with tracking and acknowledgment.

3. Estimating & Takeoff Engine

Interactive Takeoff Workspace

A professional-grade 2D takeoff workspace built on an interactive canvas for precise quantity measurement from construction drawings.

- **PDF Blueprint Rendering:** Renders multi-page PDF plan sets directly in the browser. Supports sheet navigation, renaming, duplication, and reordering.
- **Sheet Folder Organization:** Organize takeoff sheets into collapsible folder groups (Plans, Supporting Documents) with drag-and-drop reordering between groups.
- **3-Panel Layout:** Left sidebar with document tree and sheet organization, center canvas for drawing and measurement, and right panel for measurement properties and aggregation.
- **Drawing & Measurement Tools:**
  - **Select Tool (S):** Select and edit existing measurements.
  - **Pan Tool (H):** Navigate around the drawing canvas.
  - **Zoom Tool:** Zoom in/out with scroll wheel, keyboard shortcuts (+/-), or button controls.
  - **Linear Tool (L):** Measure straight-line distances between two points.
  - **Arc/Radius Tool (R):** Measure curved surfaces and arc segments.
  - **Area Tool (A):** Draw polygons to calculate enclosed areas.
  - **Volume Tool (V):** Calculate volumes from area measurements with depth input.
  - **Count Tool (C):** Place markers to count individual items (fixtures, outlets, etc.).
  - **Note Tool (N):** Pin notes and annotations directly on the drawing.
  - **Linear + Drop Tool (D):** Measure linear distances with a vertical drop component (horizontal + vertical distances combined).
  - **Pitched Linear Tool (P):** Measure along pitched/sloped surfaces with automatic pitch adjustment factor.
  - **Pitched Area Tool (T):** Calculate area on pitched surfaces (e.g., roofing) with pitch ratio multiplier.
  - **Surface Area Tool (G):** Compute 3D surface area from flat measurements using pitch conversion.
- **Calibration System:** Manual calibration by drawing a known distance, plus preset architectural, civil, and structural scales. Includes a verification mode to confirm calibration accuracy.
- **Advanced Drawing Features:**
  - Snap-to-point functionality for precise measurements.
  - Ortho-lock (Shift key) for straight horizontal/vertical lines.
  - Undo/Redo with Ctrl+Z/Ctrl+Y keyboard shortcuts.
  - Full keyboard shortcut suite with help dialog (press ? to view).
  - Real-time measurement display during drawing.
- **Addenda Overlays:** Toggle visibility of addenda overlays to compare plan revisions directly on the canvas.
- **Revision Comparison:** Side-by-side comparison of document revisions with metadata tracking.

Per-Measurement Cost Linking

Each individual measurement can be linked to cost data for real-time financial visibility during the takeoff process.

- Assign cost codes directly to individual measurements.
- Enter unit costs per measurement for instant estimated totals.
- View calculated totals (quantity x unit cost) in the measurement properties panel.
- Pitch ratio and computed values are stored and displayed for all measurement types.

Quantity Categories

Organize measurements into logical groupings that bridge the gap between takeoff quantities and estimate line items.

- Create custom categories (Concrete, Framing, Roofing, etc.) with color coding.
- Assign measurements to categories during or after the takeoff process.
- Category totals aggregate automatically across all measurements.
- Categories feed directly into the estimating engine.

Assembly Templates

Pre-built templates that auto-calculate multiple material and labor requirements from a single takeoff measurement.

- Define assemblies with multiple components, each with its own formula.
- Custom formula engine supports variables like length, area, volume, and count.
- Apply waste factors per component (e.g., 10% waste on drywall).
- Example: A "Stud Wall" assembly calculates studs, plates, drywall, tape, mud, and paint from one linear measurement.

Estimating Engine

A hierarchical line-item estimating system for complete project cost analysis.

- Create detailed estimates with line items organized by CSI division, trade, or custom groupings.
- Track five cost categories per line item: Material, Labor, Equipment, Consumables, and Subcontract.
- Automatic markup and profit calculations with configurable percentages.
- Estimate versioning with Draft, Pending Review, and Approved statuses.
- Import quantities directly from the takeoff workspace.
- Hierarchical line item builder for complex estimate structures.
- Generate professional PDF quotes and bid summaries.

Cost Codes & Labor Rates

- Comprehensive cost code system aligned with accounting standards.
- Trade-specific labor rate management with production rate benchmarks.
- Equipment rate tracking for owned and rental assets.

4. Project Management

Project Lifecycle

Projects flow through a structured lifecycle with clear status tracking at every stage.

- **Statuses:** Bid, Planning, Active, On Hold, Complete, Closed, Cancelled.
- **Conversion:** Awarded bids convert directly into projects, carrying over all scope, estimate, and document data.
- **Project Codes:** Auto-generated unique codes for easy identification and reference.

Project Detail Hub

Each project has a comprehensive detail page with specialized tabs covering every aspect of project management.

- **Overview Tab:** Quick-access dashboard showing project summary, team assignments, timeline, client information, and daily note feed.
- **Scope & Progress Tab:** Real-time tracking of installed quantities against estimated quantities for every scope of work item.
- **Daily Notes Tab:** Chronological feed of field reports, observations, and progress updates.
- **Media Tab:** Photo and document gallery with AI-powered jobsite analysis.
- **Job Information Tab:** Detailed project specifications, contract terms, and reference data.
- **Plans & Takeoff Tab:** Integrated document viewer and takeoff workspace (see Estimating & Takeoff section).
- **Estimate Tab:** Project-level estimating with hierarchical line items.
- **Procurement Tab:** Material and equipment ordering with status tracking.
- **Change Orders Tab:** Scope change management with AI-assisted drafting.
- **Time & T&M Tab:** Labor time tracking and time-and-material ticket management.
- **Labor Tab:** Crew-level labor hour tracking and analysis.
- **Receipts Tab:** Field receipt capture and expense documentation.
- **Subcontractors Tab:** Subcontractor management, assignments, and performance tracking.
- **RFIs Tab:** Request for Information management with response tracking.
- **Transmittals Tab:** Formal document transmittal tracking with receipts.
- **Submittals Tab:** Submittal processing with approval workflows.
- **Delays Tab:** Delay tracking, claims documentation, and weather evidence.
- **Inspections Tab:** Inspection scheduling, results recording, and deficiency tracking.

Project Health Score

A deterministic 0-100 score that provides an instant snapshot of project status based on weighted inputs.

- **Schedule Health (25%):** On-time performance relative to planned milestones.
- **Budget Health (25%):** Actual spending versus budgeted amounts.
- **RFI/Submittal Status (20%):** Response rates and outstanding items.
- **Change Order Impact (15%):** Volume and value of scope changes.
- **Safety Record (15%):** Incident history and compliance metrics.
- Historical snapshot trending to track health score changes over time.

Change Order Management

Full lifecycle management of scope changes with financial impact tracking.

- %ï Create change orders with detailed descriptions, justifications, and cost breakdowns.
- %ï AI Change Order Generator drafts professional documents from descriptions.
- %ï Track approval status: Pending, Approved, Rejected.
- %ï Automatically adjusts contract value and project financials upon approval.
- %ï Attach supporting documents, photos, and correspondence.

Procurement & Material Management

End-to-end material and equipment procurement tracking.

- %ï Create purchase orders with line items linked to estimate quantities.
- %ï Track order status: Planned, Ordered, Shipped, Delivered, Installed.
- %ï Material budget tracking with actual vs. budgeted consumption alerts (90% threshold warnings).
- %ï Vendor price comparison and best-pricing analysis.
- %ï AI-powered vendor pricebook extraction from PDF catalogs.

Addenda Tracking

Formal addenda lifecycle management per project with acknowledgment tracking.

- %ï Track addenda that modify original bid or project documents.
- %ï Mandatory recipient acknowledgment workflows.
- %ï Visual overlay comparison of addenda against original documents.

5. Scheduling & Field Operations

AI Scheduling

Automated schedule generation and optimization using artificial intelligence.

- %ï Generate project schedules from scope of work items and production rate data.
- %ï Interactive Gantt chart visualization with drag-and-drop adjustments.
- %ï Critical path identification and conflict detection.
- %ï Weather forecast integration for outdoor work planning.
- %ï Resource leveling based on crew availability.

Calendar

A centralized calendar view for managing deadlines, milestones, and events across all projects.

- %ï View project milestones, bid deadlines, and inspection dates.
- %ï Filter by project, team member, or event type.

Task Management

Assign and track tasks across projects and team members.

- %ï Create tasks with due dates, priorities, and assignees.
- %ï Link tasks to specific projects, bids, or scope items.
- %ï Track completion status and overdue items.

Daily Progress Tracking

Field staff record installed quantities daily, feeding real-time data into project analytics.

- %ï Log quantities installed per scope item each day.
- %ï Compare daily progress against planned production rates.
- %ï Automatic calculation of percent complete by scope item and overall project.

Delay & Claims Tracking

Comprehensive delay documentation for protecting your contractual position.

- %ï Log weather delays, owner-caused delays, and other schedule impacts.
- %ï Generate formal delay notices with supporting documentation.
- %ï Weather evidence integration with historical weather data.
- %ï Photo custody chain for legal compliance in construction disputes.

RFIs, Submittals & Transmittals

Professional document management workflows for construction communication.

- %ï **RFIs:** Create, track, and respond to Requests for Information with due date management.
- %ï **Submittals:** Process product and material submittals through approval workflows.
- %ï **Transmittals:** Formal document transmission tracking with receipt confirmation.

Inspection Management

Schedule and track inspections with results recording and deficiency management.

6. Financial Management & CFO Dashboard

CFO Dashboard

An executive-level financial command center with real-time visibility into every financial metric.

- **Contract Tracking:** Original contract value, approved change orders, and current adjusted contract amount.
- **Cost Tracking:** Committed costs, actual costs, and projected final cost.
- **Profitability Analysis:** Real-time profit margin calculations by project, trade, and cost code.
- **Cash Flow:** Current accounts receivable, accounts payable, and net cash position.
- **Variance Analysis:** Budget vs. actual comparisons with trend indicators.

Schedule of Values (SoV)

Manage progress billing and payment applications.

- Create detailed SoV breakdowns aligned with contract line items.
- Track percent complete and amounts earned per billing period.
- Retainage tracking with automatic calculations.
- Generate professional pay applications.

T&M (Time & Material) Billing

Complete Time & Material billing management for extra-contractual work.

- Create T&M daily tickets with labor, material, and equipment entries.
- Field signature capture for client authorization.
- Generate T&M invoices with detailed cost breakdowns.
- Track billing status: Draft, Submitted, Approved, Paid.

Subcontractor Payables

Manage payments to subcontractors with compliance tracking.

- Track subcontract values, change orders, and payment history.
- Monitor retainage balances and release schedules.
- Compliance verification before payment processing.

Cost Rates & Production Rates

Maintain and analyze cost and production benchmarks.

- **Labor Rates:** Trade-specific labor rates with burden calculations.
- **Equipment Rates:** Hourly, daily, and monthly rates for owned and rental equipment.
- **Production Rates:** Historical benchmarks tracking productivity per unit of work (e.g., SF of concrete poured per crew-day).
- Statistical analysis: Average, Min, Max, Median, and Standard Deviation.
- Confidence scoring based on data volume and quality.
- Cost-per-unit analysis for labor, materials, and equipment.

7. Workforce Management

Time Clock System

A native time-clock built for construction field operations.

- **GPS-Enabled Clock-In/Out:** Workers clock in and out with automatic GPS location capture.
- **Shift Management:** Configure work schedules, overtime rules, and break requirements.
- **Labor Hour Tracking:** Real-time visibility into hours worked by employee, project, and cost code.
- **Time Clock Manager:** Supervisor interface for reviewing, editing, and approving time entries.

Manager Approval Workflow

Multi-tier approval system for time entries and labor data.

- Field workers submit time entries for supervisor review.
- Supervisors review and approve or reject entries with comments.
- Approved entries flow to payroll processing.
- Flagging system for entries with overtime, missed breaks, or GPS discrepancies.

Overtime Request System

Structured process for requesting and approving overtime work.

- Workers submit overtime requests with justification.
- Managers review and approve/deny based on project needs and budget.
- Automatic overtime rate calculations based on labor rules.

Crew Management

Organize and manage field crews with assignment tracking.

- Create crews with designated foremen and team members.
- Assign crews to projects with date ranges.
- Track crew productivity and labor costs by project.
- Skills and certifications tracking per crew member.

Employee Management

Comprehensive workforce administration.

- Employee profiles with contact information, trade specialties, and certifications.
- Skills and certification tracking with expiration date alerts.
- Performance history and rating system.
- Hierarchical employee grading and rating system.

Custom KPI Management

Define, assign, and track key performance indicators for your workforce.

- Create organization-specific KPI definitions with units and target thresholds.
- Assign KPIs to individual employees with target values and period types (weekly, monthly, quarterly).
- Track achievement levels with color-coded status indicators (Green, Yellow, Red).
- Trend analysis showing improvement, decline, or stability over time.

Payroll Management

Integrated payroll processing connected to time tracking data.

- %ï Process payroll from approved time entries.
- %ï Automatic overtime calculations and rate applications.
- %ï Certified payroll report generation for prevailing wage projects.

8. CRM & Vendor Management

CRM (Customer Relationship Management)

A dedicated CRM module for managing all external business relationships.

- **Organization Types:** Track Contractors, Subcontractors, Suppliers, Vendors, Owner Representatives, and Rental Vendors as separate entity types.
- **Contact Management:** Store individual contacts within organizations with roles, responsibilities, and preferred communication methods.
- **Compliance Tracking:** Monitor insurance coverage (General Liability, Workers Comp), licenses, tax IDs, and bonding capacity with expiration alerts.
- **Interaction Logging:** Track calls, emails, meetings, and notes with interaction type tagging.
- **Lead Management:** Categorize contacts as Hot Lead, Warm Lead, or Cold Lead for sales pipeline tracking.
- **Status Management:** Track organization status as Active, Prequalified, Compliance Issue, or Inactive.

Vendor Pricing & Pricebooks

Sophisticated material pricing management for accurate estimating.

- Maintain vendor pricing catalogs with unit prices, delivery fees, and minimum order quantities.
- Track pricing by period (e.g., "2025-Q1") for historical comparison.
- AI-powered extraction of pricing from vendor PDF catalogs and Excel spreadsheets.
- Bulk pricebook import with status tracking and validation.
- Best-pricing analysis comparing multiple vendors for the same materials.
- Vendor scoring based on pricing, delivery performance, and quality.

Equipment Management

Track and manage all equipment assets.

- Equipment inventory with ownership type (Owned vs. Rental).
- Maintenance scheduling and history tracking.
- Cost tracking by project assignment.
- Rental vendor integration for equipment sourcing.

Subcontractor Management

Dedicated tools for managing subcontractor relationships and performance.

- Subcontractor scope assignments with work tracking.
- Performance scoring and historical evaluation.
- Compliance document management and verification.
- Subcontractor portal access management.

Vendor Scores & Performance

Rate and rank vendors based on real project performance data.

- Score vendors on pricing, delivery reliability, quality, and responsiveness.
- Historical performance tracking across all projects.
- Side-by-side vendor comparison for procurement decisions.

- Performance-based vendor recommendations.

Material Price Index

Track material pricing trends across vendors and over time.

- Historical price tracking for key construction materials.
- Price trend visualization and forecasting.
- Regional pricing comparison and benchmarking.
- Alert system for significant price changes.

Rental Vendor Management

Specialized management for equipment rental vendors.

- Dedicated rental vendor profiles with fleet catalogs.
- Rental rate tracking and comparison across vendors.
- Rental booking and return tracking per project.
- Cost analysis for rent-vs-own decisions.

Crew Marketplace

A marketplace for sourcing and managing specialized labor.

- Browse available crews by trade, location, and availability.
- Post crew needs and receive responses from qualified teams.
- Track crew ratings and past project performance.
- Coordinate scheduling and deployment across projects.

9. Communication & Collaboration

Unified Messaging System

A built-in messaging platform for internal team communication.

- **Team Chat (Company Channel):** Automatic organization-wide channel for announcements and general discussion.
- **Direct Messaging:** One-on-one conversations between team members.
- **Group Messaging:** Create group conversations for project teams or departments.
- **Rich Media:** Support for file attachments with message content.
- **Unread Counts & Read Receipts:** Real-time tracking of message status.
- **Organization Boundaries:** Strict tenant isolation prevents cross-organization messaging.

Client Portal

A secure, branded portal for client-facing project communication and transparency.

- Clients access project information, progress updates, and shared documents.
- Real-time chat functionality for direct communication with the project team.
- Approval workflows for change orders and pay applications.
- Document sharing with controlled access permissions.

Subcontractor Portal

A dedicated portal for subcontractor collaboration and document management.

- **Document Access:** View and respond to RFIs, Submittals, and Transmittals.
- **Comment & Upload:** Post comments and upload files to project discussions.
- **Acknowledgment Tracking:** Confirm receipt of documents and addenda.
- **Permission Levels:** Configurable access (View, Comment, Upload, Acknowledge).
- **Invitation System:** GC team members invite subcontractors via email with specific permissions.

Portal Chat

Real-time chat available in both Client and Subcontractor portals.

- Threaded conversations categorized by type (General, Billing, Technical).
- Priority levels for urgent communications.
- Visible to both the internal team and the external portal user.

Email Integration

Deep integration with email workflows beyond simple notifications.

- AI Email Assistant for classification, summarization, and draft replies.
- Integration with Gmail (Google Workspace) and Outlook (Microsoft 365).
- Automatic task creation from email content.
- Email-based notifications for bid invitations, approvals, and alerts.

Notification System

Multi-channel notifications to keep everyone informed.

- In-app real-time notifications for project activity, messages, and alerts.

- Email notifications for critical events (configurable per organization).
- Unread count badges across the application.
- Notification preferences per user.

Company Communications / Bulletins

Organization-wide announcement and communication system for company bulletins and updates.

10. Document Control & Compliance

Document Revision Management

Formal document versioning system with full revision history.

- %ï Sequential revision numbering per document.
- %ï Source tracking: Upload, Email Import, E-Signature, or System Generated.
- %ï Automatic superseding of previous revisions when new versions are uploaded.
- %ï Visual warnings when viewing older, superseded versions.
- %ï Side-by-side revision comparison with metadata diff.

Approval Workflows Framework

A generic, configurable approval engine that gates state transitions until authorized.

- %ï Applies to Change Orders, SoV Pay Applications, Timecards, and more.
- %ï Role-based routing (e.g., Superintendent for field approvals, Office Manager for billing).
- %ï Dedicated approval inbox for managers to review and act on pending items.
- %ï Audit trail for all approval/rejection actions with timestamps and comments.

Audit Logging

Comprehensive, immutable audit trail for all system activity.

- %ï Captures all Create, Update, Delete, Approve, and Reject actions.
- %ï Records performing user, timestamp, IP address, user agent, and detailed change diff (old value vs. new value).
- %ï AI-specific audit logging for all AI-generated content and recommendations.
- %ï Exportable in CSV and JSON formats for external compliance reviews.

Photo Custody Chain

Legal-grade photo documentation for construction disputes and claims.

- %ï Maintains metadata and hash verification for photographic evidence.
- %ï Ensures chain of custody integrity from capture to presentation.
- %ï GPS and timestamp verification on captured images.

Data Retention Policies

Automated data lifecycle management for compliance.

- %ï Configure retention periods by data type (e.g., Audit Logs: 7 years, Messages: 2 years).
- %ï Supported actions: Archive or Delete upon expiration.
- %ï Dry-run preview before execution to see affected record counts.
- %ï Automated enforcement on configured schedules.

GDPR & Privacy Compliance

Enterprise-grade privacy tooling for regulatory compliance.

- %ï Records of Processing Activities (ROPA) management.
- %ï Data Subject Access Request (DSAR) handling workflow.
- %ï Right-to-erasure implementation with configurable scope.

- Privacy impact assessment tools.

11. Website Builder & Social Media

Website Builder

A complete, no-code website builder that allows construction companies to create and manage professional websites directly from their SmartManageAI dashboard.

Page Builder & Editor

- **Component-Based Design:** Build pages using modular components: Hero sections, About blocks, Services listings, Project Galleries, Testimonials, Contact Forms, and more.
- **Drag-and-Drop Reordering:** Visually reorder page sections with drag-and-drop.
- **Device Previews:** Real-time preview across Desktop, Tablet, and Mobile viewports.
- **Global Components:** Save sections as reusable components across multiple pages.
- **Version History:** Track and revert changes with full page version history.
- **Text Styling:** Rich text editor with font, color, and layout controls.
- **Media Library:** Integrated media management for images and assets.

AI Website Assistant

- **AI Design Consultant:** Chat-based assistant that suggests layouts, color schemes, and content based on your trade and brand.
- **AI Page Generation:** Automatically generates complete pages (Home, About, Services, Safety, Contact) with SEO-optimized content tailored to your construction specialty.
- **Context-Aware Content:** Pulls from your Scope of Work data to generate accurate service descriptions.

Publishing & Domain Management

- **Site Status Management:** Draft, Published, or Maintenance modes.
- **Staging Preview Links:** Generate secure, token-protected preview URLs for sharing draft sites with stakeholders before going live.
- **Custom Domain Mapping:** Connect your own domain with DNS verification tracking.
- **Deployment Logging:** Full audit trail of who published what and when.
- **Branding Controls:** Custom colors, logos, and fonts to match your company brand.

Social Media Management

A comprehensive social media suite for managing your construction company's online presence across all major platforms.

Supported Platforms

Facebook, Instagram, LinkedIn, X (Twitter), TikTok, YouTube, and Pinterest.

Account Management

- Connect and manage business accounts for each platform.
- Encrypted token storage for security.
- Connection health monitoring and automatic status tracking.

Content Creation & Scheduling

- **Social Calendar:** Monthly and weekly calendar views for visual content planning.
- **Post Scheduling:** Schedule posts for specific dates and times, or publish immediately.

- **Recurring Posts:** Set up daily, weekly, or monthly recurring content.
- **AI Content Generation:** Generate platform-specific content tailored to character limits and style conventions (professional for LinkedIn, visual for Instagram, etc.).
- **Hashtag Management:** Add and manage hashtags per post.
- **Campaign Tags:** Organize posts by marketing campaign.
- **UTM Parameters:** Automatic UTM tracking for measuring traffic from social posts.

Approval Workflow

Posts follow a structured lifecycle: Draft, Pending Approval, Approved, Scheduled, Queued, Posted (or Failed).

Analytics & Reporting

- Track total posts, scheduled vs. published, and failure rates.
- Platform-by-platform activity breakdown.
- Engagement metrics and campaign performance tracking.

12. Dashboard & Analytics

Customizable Dashboard

A highly flexible, widget-based dashboard that each user can personalize to their role and preferences.

Layout Customization

- **Drag-and-Drop Grid:** Arrange widgets in a responsive grid layout.
- **Section Management:** Organize widgets into labeled sections with configurable column counts (1-4 columns).
- **Edit Mode:** Add, remove, duplicate, and reorder widgets and sections with a visual editor.
- **Role-Based Defaults:** Pre-configured layouts for different roles (Admin, Manager, Superintendent, Estimator, etc.).

Widget Library (20+ Widgets)

- **KPI Summary Card:** Display any metric with trend comparison, sparklines, and color-coded status.
- **Trend Chart:** Configurable line or bar charts for time-series data with customizable date ranges.
- **Cost Code Breakdown:** Pie charts for cost distribution and bar charts for Budget vs. Actual comparison.
- **Workforce Snapshot:** Real-time metrics on active workers, overtime hours, and labor costs.
- **Cash Position:** Financial health overview showing AR, AP, Payroll, and Net Cash.
- **SOV Progress:** Schedule of Values billing progress and retainage tracking.
- **Bid Pipeline:** Visualization of active bids, win rates, and total pipeline value.
- **Schedule Impact:** Project timeline analysis with milestone tracking.
- **AI Insights:** AI-generated observations about project health and recommendations.
- **Active Projects:** Summary of all current projects with status indicators.
- **Active Bids:** Upcoming bid deadlines and opportunity tracking.
- **Calendar Events:** Upcoming meetings, inspections, and deadlines.
- **Messages & Notifications:** Recent communications and alert badges.
- **Quick Stats:** Configurable metric tiles for at-a-glance performance data.
- **Outstanding Items:** Overdue RFIs, submittals, and action items.
- **Project Assignments:** Current project assignments for the logged-in user.
- **Crew Overview:** Active crew deployment and availability.

Theme Customization

- Color palette customization with light and dark mode support.
- Custom hex color configuration for brand alignment.
- WCAG AA compliant contrast ratios (4.5:1 minimum) for all theme colors.
- Preset themes and custom palette creation.

Production Rate Analytics

Deep analytics for understanding and benchmarking crew productivity.

- Historical production rate tracking by scope of work type (e.g., SF of drywall installed per crew-day).
- Statistical analysis: Average, Min, Max, Median, and Standard Deviation.
- Labor, material, and equipment cost-per-unit analysis.
- Confidence scoring based on data volume (High, Medium, Low).

%ï Cross-project benchmarking for continuous improvement.

Executive Digest

High-level reporting page for management-level summaries of business performance across all projects.

What-If Simulator

Scenario planning tool for evaluating the financial impact of potential decisions before committing.

13. Integrations & Integration Hub

Integration Hub Architecture

SmartManageAI features a provider-agnostic integration framework designed for reliability and extensibility.

- **Centralized Connection Management:** All third-party connections are managed through a single, unified interface.
- **OAuth Token Management:** Secure storage and automatic refresh of access tokens for all connected services.
- **Event Outbox System:** A reliable queuing system that ensures events are delivered to external services even during temporary outages.
- **Webhook Subscriptions:** Outgoing webhook management with payload signing for security.
- **Integration Audit Log:** Comprehensive logging of all connection events, sync operations, and failures.

QuickBooks Online (Full Integration)

The most comprehensive integration, providing two-way financial data synchronization.

- **OAuth 2.0 Connection:** Secure authorization flow with automatic token refresh.
- **8 Sync Operations:**
  - Chart of Accounts synchronization.
  - Customer records sync.
  - Vendor records sync.
  - Invoice creation and updates.
  - Bill management.
  - Purchase Order sync.
  - Payroll data export.
  - Payment recording.
- **Cost Code Mapping:** Map SmartManageAI cost codes to QBO account categories.
- **Automatic Sync Scheduling:** Configure sync intervals for hands-free financial data flow.
- **Manual Sync:** On-demand sync for immediate data updates.

Google Workspace

- **Gmail Integration:** Read-only access to email for the AI Email Assistant.
- **Google Calendar:** Two-way calendar synchronization for project milestones and meetings.

Microsoft 365

- **Outlook Mail:** Email integration via Microsoft Graph API.
- **Microsoft Calendar:** Calendar synchronization for scheduling and deadline tracking.

DocuSign (E-Signature)

- E-signature workflows for contracts, change orders, and subcontract agreements.
- Document routing and status tracking.
- Signed document archival with revision tracking.

Zapier

- Webhook-based automation for connecting SmartManageAI events to 5,000+ external applications.

- Event dispatching for project updates, bid activity, and approval actions.
- Custom workflow automation without coding.

OpenAI (GPT-4o & GPT-4o Vision)

- Powers all AI features throughout the platform.
- Vision capabilities for blueprint analysis and jobsite photo assessment.
- Text generation for document drafting, analysis, and recommendations.
- Governed by the AI Governance Dashboard for cost and compliance monitoring.

Resend (Email Service)

- Transactional email delivery for all system notifications.
- Password reset emails, bid invitations, and approval notifications.
- Delivery tracking and bounce handling.

14. Mobile & Offline Capabilities

Progressive Web App (PWA)

SmartManageAI is built as a full Progressive Web App, installable on any mobile device.

- %ï Install directly from the browser — no app store required.
- %ï Full-screen standalone mode for a native app experience.
- %ï Home screen icon with branded splash screen.
- %ï Responsive design optimized for phone, tablet, and desktop.

Offline Mode & Data Sync

Designed for construction sites where internet connectivity is unreliable.

- %ï **Offline Queue:** When connectivity is lost, actions (Time Entries, Daily Notes, Photos, Voice Logs) are stored locally in an offline queue.
- %ï **Automatic Sync:** When connectivity returns, queued actions are automatically synchronized with the server.
- %ï **Conflict Resolution:** Detects conflicts when the server has newer data than the local device. Provides a clear UI for choosing "Local Wins" vs. "Server Wins" resolution.
- %ï **Resilient Upload:** Background retry with exponential backoff for large file uploads (photos, plans).
- %ï **Sync Status Indicator:** Real-time connectivity status display with pending change counts.

Mobile-Optimized Features

- %ï GPS-enabled time clock for field workers.
- %ï Photo capture with automatic metadata (GPS, timestamp).
- %ï Voice log recording for hands-free field notes.
- %ï Touch-optimized interfaces for field data entry.

15. Security & Enterprise Features

Authentication

- **JWT-Based Authentication:** Secure JSON Web Tokens stored in httpOnly cookies, protecting against XSS attacks.
- **Password Security:** Bcrypt hashing with configurable complexity requirements.
- **OAuth Social Login:** Sign in with Google or Apple for streamlined access.
- **CSRF Protection:** Cross-Site Request Forgery prevention on all authenticated endpoints.
- **Email Verification:** Required email verification for new account registration.

SSO / SAML 2.0

Enterprise-grade Single Sign-On for large organizations.

- Support for major Identity Providers: Okta, Azure AD, OneLogin, and any SAML 2.0 compliant IdP.
- Metadata URL support for automatic configuration.
- Certificate validation for secure assertion verification.
- "Enforce SSO" mode to disable standard password login for maximum security.

Session Security

Advanced session management and control.

- **Token Versioning:** Changing passwords or administrative revocation immediately invalidates all active sessions globally.
- **Configurable Idle Timeout:** Default 8-hour session expiry with organizational override.
- **Concurrent Session Limits:** Maximum of 5 simultaneous sessions per user (configurable).
- **Session Revocation:** Administrators can revoke all sessions for any user instantly.

Role-Based Access Control (RBAC)

A sophisticated hierarchical permission system with 18 distinct roles.

- **Platform Level:** SUPER_ADMIN for platform-wide management.
- **Organization Level:** ADMIN, MANAGER, OFFICE_MANAGER for organizational leadership.
- **Operations Level:** SUPERINTENDENT, FOREMAN, PROJECT_MANAGER for project operations.
- **Specialty Roles:** ESTIMATOR, ACCOUNTANT, SAFETY_OFFICER for functional expertise.
- **Field Roles:** FIELD_ENGINEER, HOURLY_EMPLOYEE for field workers.
- **External Roles:** VENDOR, CLIENT for portal access.
- **Enterprise Roles:** HUSTLER for multi-organization management.
- **Triple-Layer Enforcement:**
  1. Route-level middleware blocking unauthorized API access.
  2. Feature-level gating based on subscription tier and feature flags.
  3. Data-level organization scoping on all database queries.

Multi-Tenant Architecture

Complete data isolation between organizations.

- Every database query is scoped to the requesting user's organization.
- Cross-tenant data access is physically prevented at the query level.

- Organization-specific configuration, branding, and feature toggles.

The Hustler Role (Multi-Organization Management)

A unique enterprise feature for business owners who manage multiple companies.

- Single login to manage multiple organizations.
- Unified inbox and dashboard spanning all managed organizations.
- Delegation mode for assuming specific roles within sub-organizations.
- Cross-organization analytics and benchmarking.

Rate Limiting

Protection against abuse and automated attacks.

- Sensitive endpoint protection (authentication, billing) with rate limiting.
- Brute-force prevention on login endpoints.
- API rate limits per user and organization.

Feature Flags

Granular control over feature availability.

- Enable or disable specific modules per organization.
- Managed by SUPER_ADMIN for platform-wide control.
- Supports gradual feature rollouts and A/B testing.

16. Subscription Plans & Billing

Four-Tier Subscription Model

Trial / Starter (Free)

Core project management features to get started.

- Basic project tracking and document management.
- Time clock and daily notes.
- Limited team size.

Growth ($229/month)

For growing construction companies that need deeper operational insight.

- Everything in Starter, plus:
- Production rate tracking and analytics.
- Advanced reporting and dashboards.
- Expanded team size and project limits.

Scale ($579/month)

For established companies needing enterprise security and portfolio management.

- Everything in Growth, plus:
- SSO/SAML 2.0 authentication.
- Audit log export.
- Portfolio dashboards and cross-project analytics.
- Advanced approval workflows.

Hustler ($1,199/month)

For multi-company operators and enterprise construction groups.

- Everything in Scale, plus:
- Multi-organization management.
- Cross-organization analytics and benchmarking.
- Unified inbox across all companies.
- White-glove onboarding support.

Modular Add-Ons

Extend any plan with specialized capabilities.

- **AI Project Manager Add-On:** Full AI PM and CFO analytics.
- **AI CFO Add-On:** Advanced financial intelligence and forecasting.
- **QuickBooks Integration Add-On:** Full QBO sync with automatic scheduling.
- Additional add-ons for specific industry needs.

Billing Management

- Self-service subscription management from the platform.
- Usage tracking and entitlement enforcement.
- Transparent pricing with no hidden fees.

17. Platform Administration

Super Admin Dashboard

Platform-level administration for managing the entire SmartManageAI ecosystem.

- **Organization Management:** View, create, and manage all tenant organizations with detailed organization profiles.
- **User Management:** Cross-organization user administration and account oversight.
- **Platform Analytics:** Usage metrics, growth trends, active user counts, and system health monitoring.
- **Platform Audit Dashboard:** Cross-tenant audit log visibility for compliance and security monitoring.
- **Subscription Management:** View and manage subscription tiers, add-ons, and revenue tracking.
- **Support Tools:** User impersonation capability for troubleshooting issues while maintaining audit trail.
- **Feature Flag Management:** Enable/disable features globally or per organization for staged rollouts.

Company Health Dashboard

Organization-level health monitoring and operational metrics.

- Overall business health score based on active projects, revenue, and team utilization.
- Key operational metrics at a glance: active projects, open bids, labor utilization, and financial standing.
- Trend tracking for month-over-month business performance.

Hustler Multi-Organization Management

The Hustler tier provides a unique command center for business owners managing multiple companies.

- **Hustler Dashboard:** Unified overview across all managed organizations with cross-company KPIs.
- **Organization Detail View:** Drill into any managed organization's performance, financials, and operations.
- **Unified Inbox:** Single inbox aggregating messages, notifications, and action items from all organizations.
- **Benchmarking:** Compare performance metrics across organizations to identify best practices and underperformers.
- **Delegation Mode:** Assume specific roles within sub-organizations without switching accounts.

Organization Administration

Tools for organization-level administrators.

- **Company Settings:** Business information, branding, trade configuration, and system preferences.
- **User & Role Management:** Invite team members, assign roles, and manage permissions.
- **Scope Settings:** Configure trade-specific scope of work categories and items.
- **Scope Library:** Browse and import from a catalog of 2,946 industry-standard scope items across 230 trades.
- **Crew & Skills Management:** Manage crews, worker assignments, and certification tracking.
- **Report Lock Management:** Lock daily reports for specific dates to prevent retroactive changes.

AI Onboarding

An intelligent onboarding system that configures the platform for your specific trade.

- **Trade Discovery:** AI researches your construction specialty to pre-configure relevant scope items, materials, formulas, and calculators.

- **Contractor Onboarding Library:** A comprehensive reference database powering the onboarding flow:
  - 230 trades across 17 categories.
  - 2,946 pre-built scope items.
  - 126 material families organized into 17 groups.
  - 39 industry-standard formulas.
  - 982 trade components with variant expansions.
  - Standardized line item grammar with naming rules and UOM codes.
- **Guided Walkthrough:** Step-by-step setup wizard that personalizes the platform to your business.

18. Construction Calculators & Tools

Built-In Calculators

A suite of trade-specific calculators accessible directly from the platform.

- **Roofing Calculator:** Squares, bundles, ridge cap, underlayment, and starter strip calculations with pitch adjustments.
- **Drywall Calculator:** Sheets, joint compound, tape, screws, and corner bead from room dimensions.
- **Masonry Calculator:** Block counts, mortar, sand, and rebar for wall construction.
- **Paint Calculator:** Gallons, primer, and coverage area with multiple coat support.
- **Tiling Calculator:** Tile count, thin-set, grout, and spacer quantities.
- **Lumber Calculator:** Board feet, linear feet, and piece counts for framing.
- **Aggregates Calculator:** Cubic yards of gravel, sand, and stone for base and fill work.
- **Geometry Calculator:** Area and volume calculations for common shapes (rectangle, circle, triangle, trapezoid).
- **Markup Calculator:** Calculate selling price from cost with desired markup or margin percentages.
- **Profit Margin Calculator:** Analyze profit margins across different pricing scenarios.
- **Labor Cost Estimator:** Total labor cost from crew size, hours, and rates.
- **Material Estimator:** Generic material quantity calculator with waste factor.
- **Unit Converter:** Convert between imperial and metric units for all construction measurements.

Voice Logs

Hands-free field documentation through voice recording.

- Record voice memos from the field.
- Attach to projects, daily notes, or specific scope items.
- Timestamped and GPS-tagged for documentation accuracy.

Issue Captures

Quick documentation of field issues and deficiencies.

- Photo-based issue reporting with annotations.
- Priority and category tagging.
- Assignment and resolution tracking.

Summary of Key Platform Highlights

Category	Highlights
AI Intelligence	12+ AI-powered features including Vision-based Auto-Takeoff, Project Manager, CFO Analyst, Scheduling Assistant, and Email Classifier
Trade Support	230+ construction trades across 17 categories with trade-specific configuration
Measurement Types	10 measurement tools including pitched and surface area calculations
Integrations	QuickBooks Online, Google Workspace, Microsoft 365, DocuSign, Zapier, and OpenAI
User Roles	18 distinct roles with triple-layer RBAC enforcement
Subscription Tiers	4-tier model (Starter through Hustler) with modular add-ons
Security	JWT auth, SAML 2.0 SSO, session management, GDPR compliance, and audit logging
Mobile	Full PWA with offline mode, GPS time clock, and field photo capture
Scope Library	2,946 pre-built scope items, 126 material families, and 39 formulas
Collaboration	Team messaging, client portal, subcontractor portal, and bid network
Website & Social	No-code website builder with AI wizard plus 7-platform social media management
Marketplace	Vendor scoring, material price index, crew marketplace, and rental vendor management
Calculators	13+ trade-specific construction calculators built in
Dashboards	20+ customizable widgets with drag-and-drop layout editor
Document Control	Formal revisioning, approval workflows, addenda tracking, and photo custody
Multi-Org (Hustler)	Unified dashboard, cross-org benchmarking, delegation mode, and unified inbox
Platform Admin	Organization management, platform analytics, audit dashboard, and support tools

SmartManageAI — The intelligent construction management platform that works as